

Sofija Kaluderovic

 skaluderovic |  sofijakaluderovic |  sofijakaludjerovic@gmail.com |  +38267863461

MSc Data Science graduate from **University College London (UCL)** with a strong background in Economics & Finance and quantitative analysis. Experienced in cleaning, validating, and modelling real-world data through an industry-sponsored data science project, as well as in statistical modelling, machine learning, and time-series analysis using Python and R. Particularly interested in applying data science and quantitative methods to financial markets and data-driven decision-making.

EDUCATION

- 09/2025 - 09/2026 MSc Data Science at **University College London (UCL)**
London, UK
- Current average: **76.8%** across six released modules, with all marks at Distinction level (70%+)
 - Selected modules: Stochastic Methods in Finance, Applied Machine Learning, Statistical Computing, Inference at Scale
 - Awarded the national **Scholarship for Excellence in Higher Education** (€25,000; one of 32 recipients nationwide); also recipient of the **Capital City of Podgorica Scholarship** for high-achieving students
 - Elected **Course Representative** for the MSc Data Science 2025/26 cohort, representing students and contributing to programme improvements
- 09/2022 - 05/2025 BA, double major in Economics & Finance and International Affairs at **John Cabot University (JCU)**
Rome, Italy
- **3.92/4.0 GPA**, *Summa Cum Laude*
 - Minor in Business Administration; selected quantitative coursework: Econometrics, Mathematical Statistics, Applied Data Analytics, Linear Algebra, Calculus II
 - Recipient of the **Certificate of Academic Excellence in Finance and Accounting**
 - Academic Representative in Student Government and Secretary of the International Relations Society

PROFESSIONAL EXPERIENCE

- Industry Data Science Project with Dogs Trust – UCL MSc Dissertation** 06/2026 – 09/2026, London, UK
Selected through a competitive application and interview process for an industry-sponsored project with Dogs Trust, the UK's largest dog welfare charity. Cleaned, merged, and validated multiple real-world datasets, including one containing over 30,000 records. Conducted statistical analyses in R, including mixed-effects modelling and inter-rater reliability, correlation, longitudinal, and sensitivity analyses, and communicated findings to Dogs Trust and UCL stakeholders.
- Resident Assistant – Housing Office, JCU** 01/2023 – 05/2025, Rome, Italy
Managed student residence operations, resolved complex issues through practical problem-solving, documented incidents, organised events, and collaborated with an international team, including during time-sensitive emergency situations.
- Organising Committee – Eurasia Business & Economics Society Conference, JCU** 04/2025, Rome, Italy
Co-organised the conference, coordinated sessions and logistics, facilitated research presentations, and communicated with speakers, organisers, and attendees.
- Conference Supervisor – Splendid Conference & SPA Resort 5*** 05/2023 – 07/2023, Budva, Montenegro
Supervised conference operations, coordinated logistics, and worked closely with hotel management and cross-functional teams to ensure the smooth delivery of events.

TECHNICAL SKILLS

- Programming:** Python (pandas, NumPy, scikit-learn), R, SQL, Stata
Tools: GitHub, Microsoft Excel
Methods: Machine Learning, Statistical Modelling, Data Cleaning & Validation, Time Series Analysis & Forecasting, NLP & Generative AI (LLM Fine-Tuning), Model Evaluation, Regression Analysis, Data Visualisation
Certification: Bloomberg Market Concepts (BMC)

LANGUAGES

Serbian (Native) • **Russian** (Native) • **English** (Fluent) • **Italian** (Fluent)

SELECTED PROJECTS

- Chakma-English Machine Translation – NLP & LLM Fine-Tuning (UCL Group Project)**
Developed, as part of a UCL team, the first Chakma-English word-level translation system for Chakma, an endangered Indo-Aryan language. Curated 20,000+ Chakma-English word pairs verified by native speakers. Fine-tuned LLaMA 3.1 8B and Gemma 3 4B using QLoRA and evaluated performance across multiple translation metrics, with both models improving over their base versions.
- PM2.5 & Life Expectancy (US) – R, Geospatial Analysis**
Integrated and cleaned multiple US air-quality, life-expectancy, and geographic datasets; performed geospatial analysis, data visualisation, and regression modelling to examine the association between PM2.5 exposure and life expectancy. Reproducible analysis is available on GitHub.

ADDITIONAL ACHIEVEMENTS & INTERESTS

Member of **Mensa** since 2020; winner of the **JCU Chess Tournament**; coordinator of the **TEDx “Third Self”** event; published creative writer, with my fiction, poetry, and short stories featured in two Montenegrin anthologies.